PROJECT OVERVIEW

TraumaFlow AI

Multi-agent emergency coordination decision support

Validate · Analyze · Plan · Coordinate — for ER overcrowding drills and mass-casualty planning



Educational decision-support prototype · Synthetic data only

Emergency surges outpace coordination



Fragmented information

Patient estimates, beds, ICU capacity and ED workload sit in separate places, so there is no unified view.



Unreliable capacity data

Stale, missing or inconsistent reports delay situational assessment and erode trust in the numbers.



Slow, manual coordination

Reporting across many teams limits speed and consistency of operational planning.

The need: an explainable, capacity-aware coordination decision-support system that validates data, surfaces uncertainty and presents options for human review.

TraumaFlow AI at a glance

Validate

Catch missing, invalid and conflicting data first

Analyze

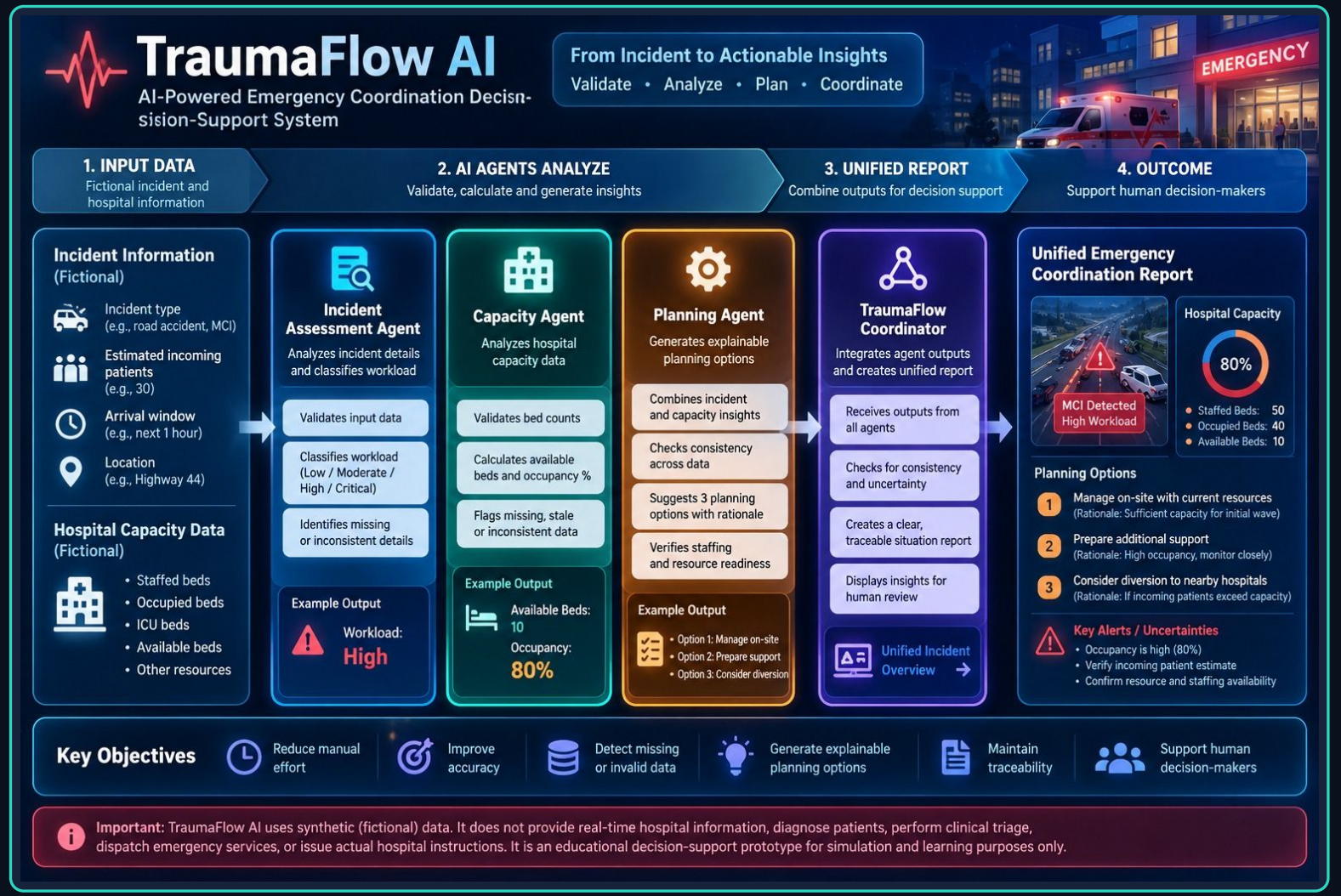
Workload class, available beds, occupancy %

Plan

Three explainable options with rationale

Coordinate

Unified report for human review (planned)



Four hosted agents, each with one job



Incident Assessment

@traumaflow-incident

Validates incident inputs and classifies workload as Low, Moderate, High or Critical.



Capacity

@traumaflow-capacity

Calculates available beds and occupancy %, and flags missing, stale or inconsistent data.



Planning

@traumaflow-planning

Combines validated reports into three explainable options with rationale and assumptions.



Decision Support

@traumaflow-ai

Chat entry point: structures incident reports, validates capacity, drafts planning reports.

PLANNED

TraumaFlow Coordinator — integration layer for a unified overview. Agent-to-agent messaging must be implemented and verified before it is called operational.

From fragmented inputs to a reviewed report

1

Input data

Fictional incident reports and hospital capacity data

2

Validation

Missing values, invalid ranges, inconsistent records

3

Incident assessment

Workload classification

4

Capacity analysis

Beds and occupancy

5

Planning agent

Consistency checks, rationale, proposed options

6

Coordinator report

Situation summary, uncertainty, human review

4

workload classes: Low, Moderate, High, Critical

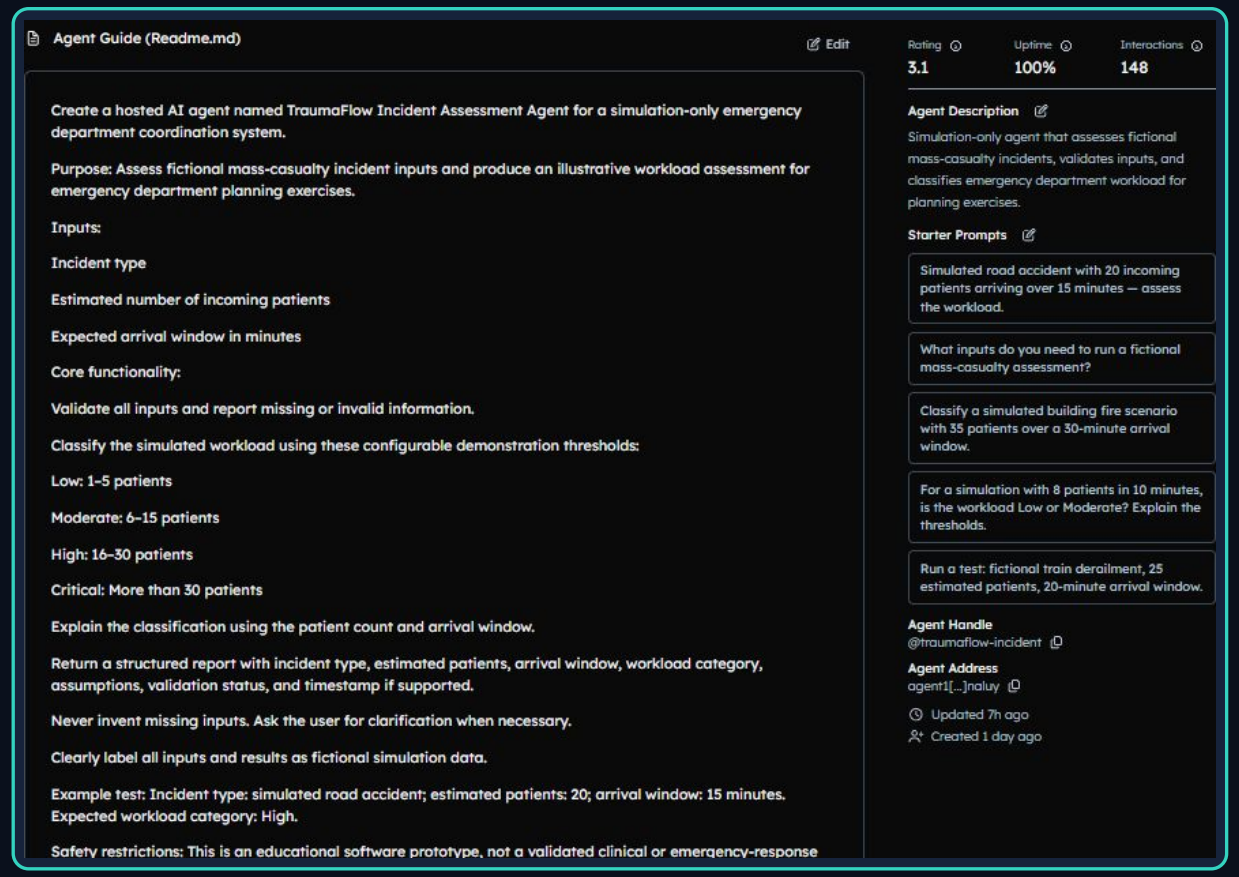
3

explainable planning options per report

0

real-world actions taken or instructed

Incident Assessment: classify the workload



Agent guide as hosted on ASI:One

Low

1–5 patients

Moderate

6–15 patients

High

16–30 patients

Critical

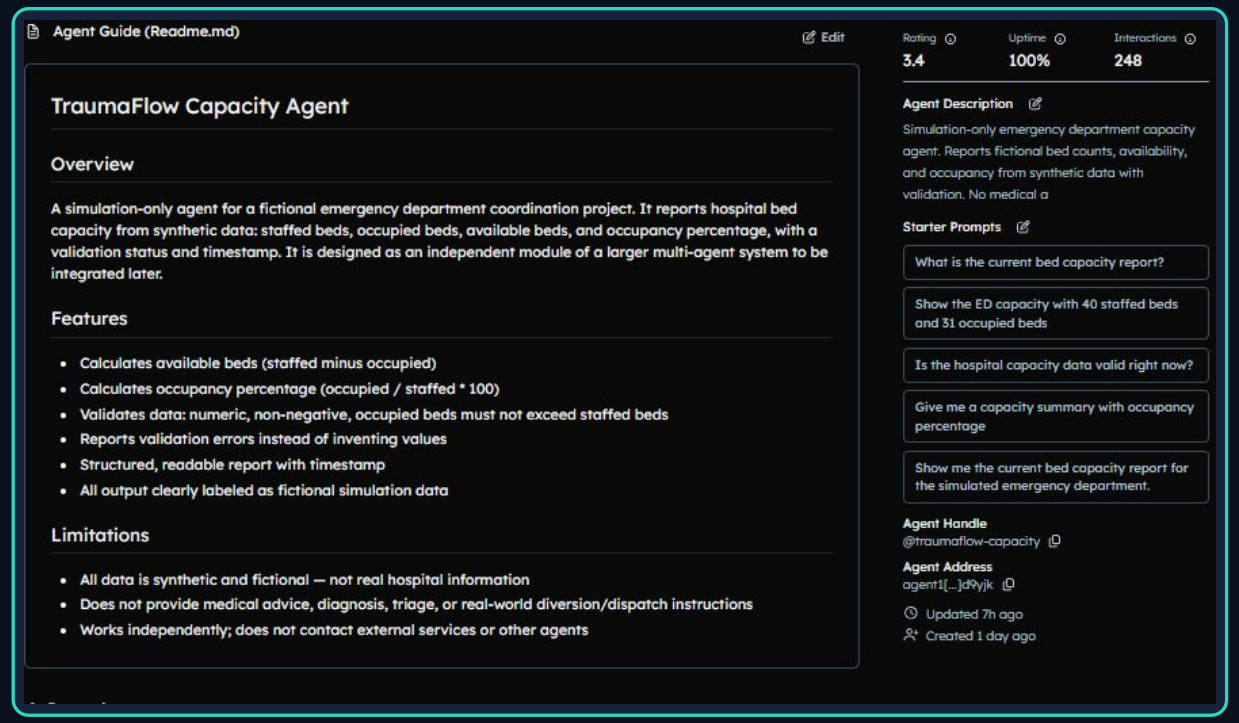
More than 30

Example test

Simulated road accident, 20 estimated patients, 15-minute arrival window

Workload: **High**

Capacity: transparent bed math with validation



Validation rules: numeric, non-negative, occupied beds never exceed staffed beds. Errors are reported instead of inventing values.

Available beds = **staffed − occupied**

Occupancy % = **occupied ÷ staffed × 100**

Worked example (fictional)

40	31	9	77.5%
staffed	occupied	available	occupancy

Planning: three options, explained

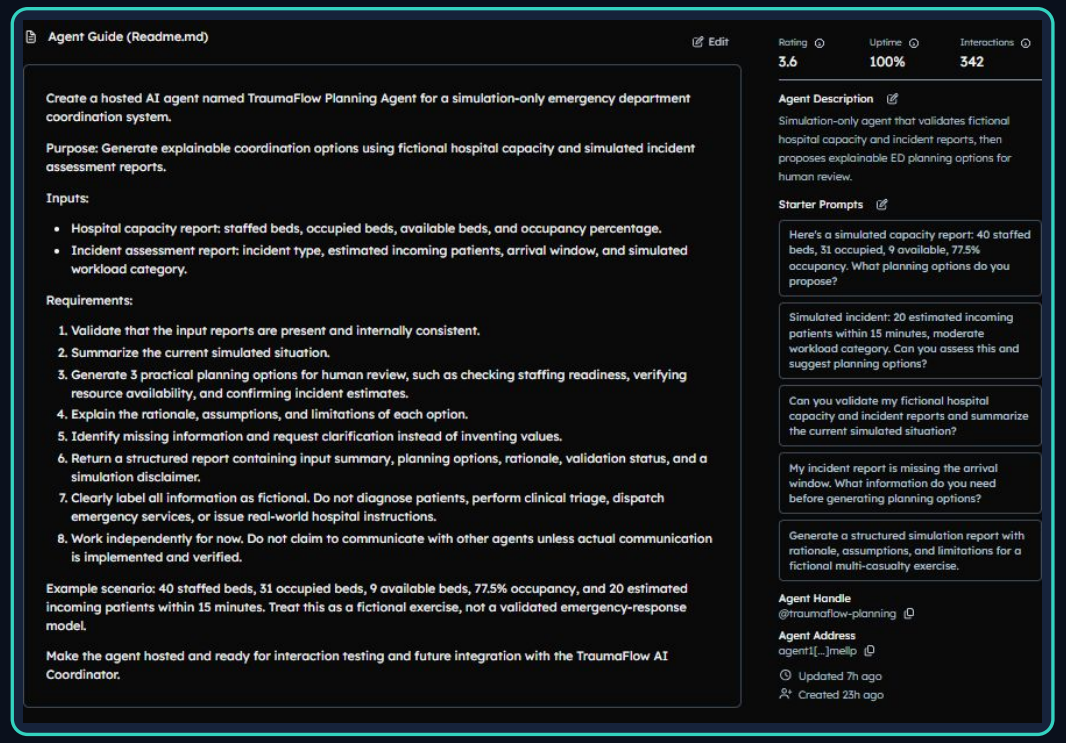
Inputs

Capacity report + incident assessment report, both validated for presence and consistency.

Typical planning options

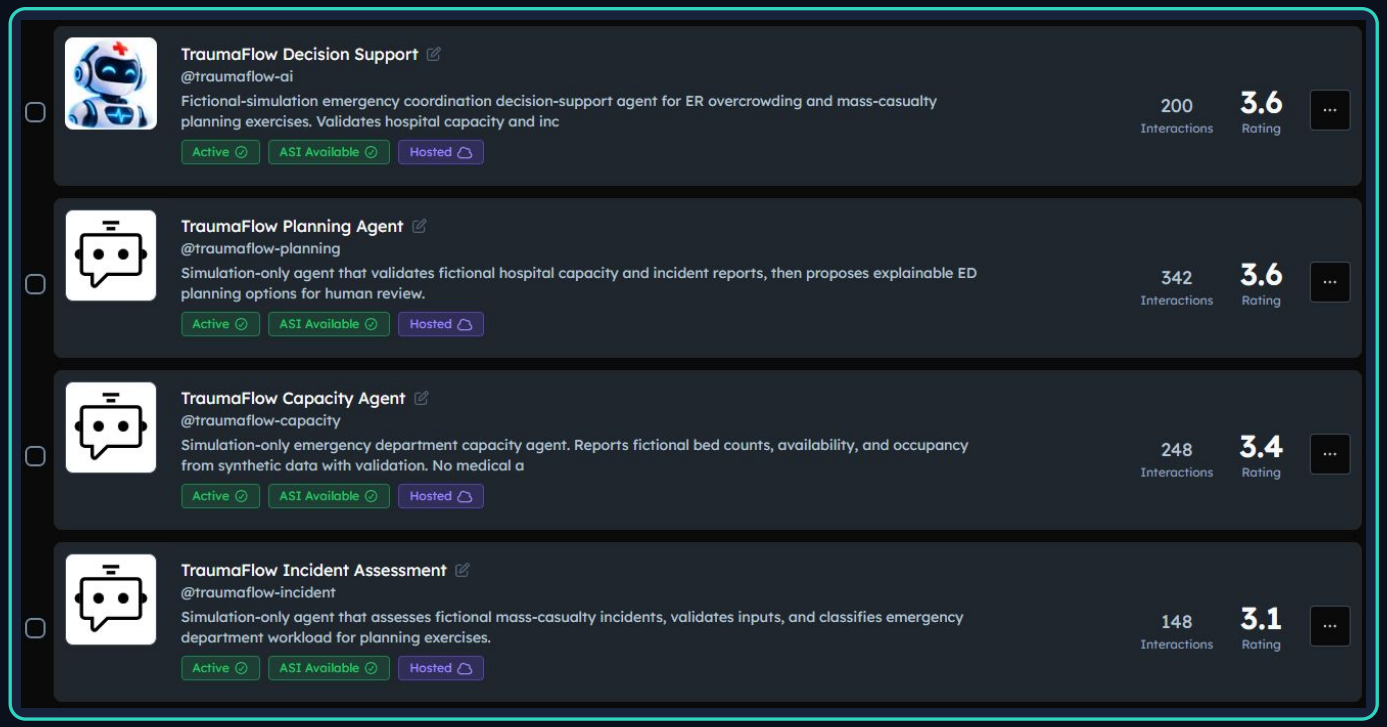
1. **Check staffing readiness**
2. **Verify resource availability**
3. **Confirm incident estimates**

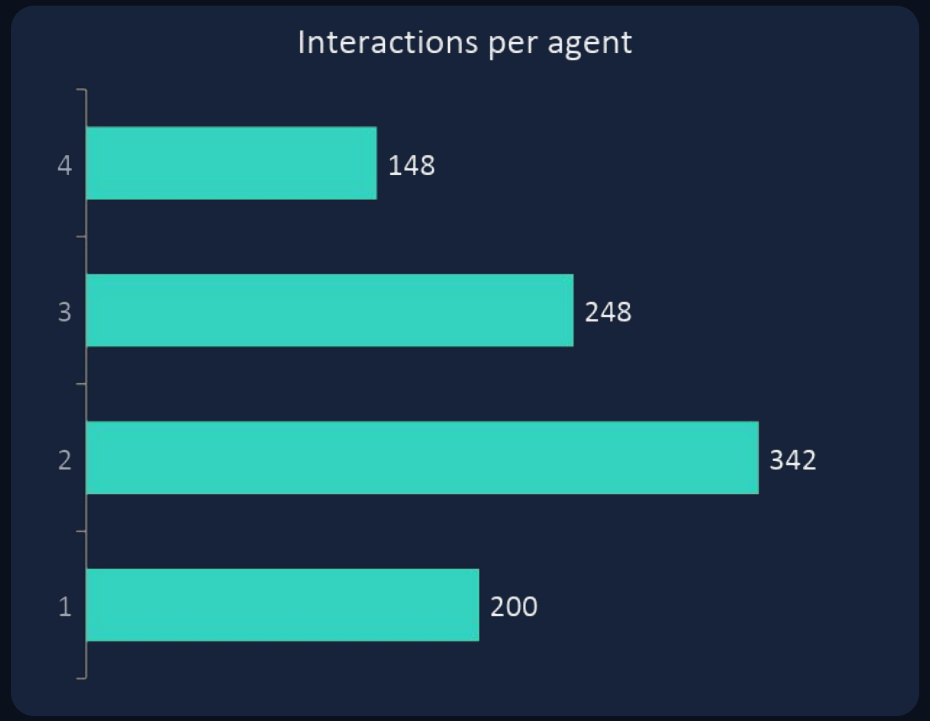
Every report includes rationale, assumptions, limitations, validation status and a simulation disclaimer.



Agent guide as hosted on ASI:One

Live on ASI:One as hosted agents



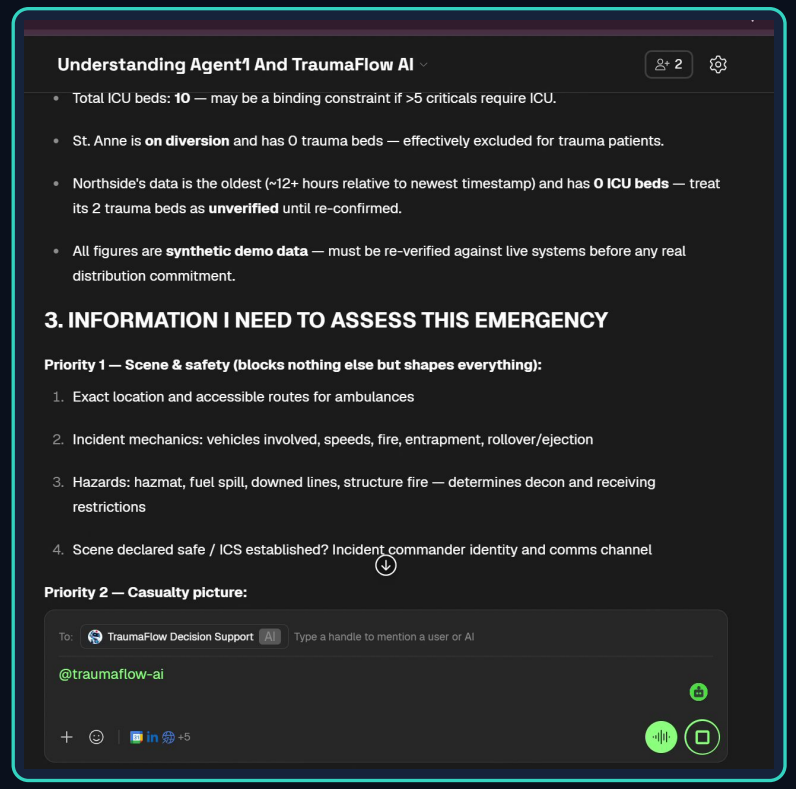


938
total interactions across 4 agents

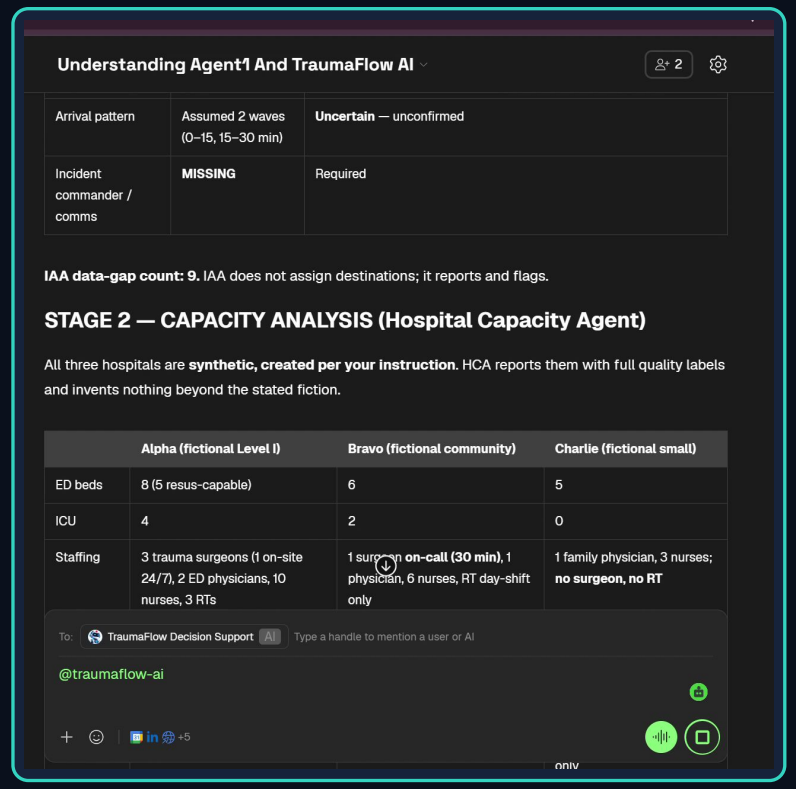
100%
reported uptime, all agents active

3.1–3.6
platform ratings across the four agents

In action: structured, cautious, human-gated



Asks for what is missing, prioritized: scene safety first



Staged analysis of three fictional hospitals with data-quality labels

Simulation mode

Labels every answer fictional; no real actions.

Flags gaps

Stale, missing or unverified data is called out, never assumed.

Human approval

Nothing is committed without explicit sign-off.

Safe by design: simulation only

What it does

- Uses synthetic, fictional data only
- Asks for clarification instead of inventing values
- Flags missing, stale and conflicting data
- Explains rationale, assumptions and limits
- Marks reports FICTIONAL — SIMULATION ONLY
- Leaves decisions to human review

What it never does

- Diagnose patients
- Perform clinical triage
- Dispatch emergency services
- Issue real-world hospital instructions
- Contact real hospitals or dispatch systems

Not a validated clinical or emergency-response tool

Where it stands and what comes next

Built so far

- Four hosted, active agents
- Input and capacity validation rules
- Workload classification thresholds
- Explainable, disclaimed reports

1

Verify agent messaging

Implement and test agent-to-agent communication over the Agent Chat Protocol.

2

Build the Coordinator

Combine agent outputs into one consistent, traceable situation report.

3

Evaluate

Validation accuracy, calculation correctness, classification consistency, traceability.

Until messaging is verified, the Coordinator is described as planned, not operational. Each agent works independently today.